

Quantum-generated singlet interactions and canonical matching

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Result and physical scope

The shared-singlet completion admits a non-Gaussian extension containing every singlet self and portal structure generated by its quark determinant, while preserving the retained stable local CP-breaking vacuum and canonical matching. The common-singlet mass relation extends beyond a quadratic singlet potential: it is an exact classical functional relation on every isolated positive-curvature stationary branch. This closes a specific assumption in the preceding Gaussian construction without claiming full finite one-loop EFT matching or all-loop mass-phase protection.

The same calculation gives a precise mechanism-selection result. The entire one-loop quark effective potential has zero orientation torque along independent fixed-eigenvalue source rotations. It can affect radial masses and compare vacua with different spectra, but it cannot choose the relative source eigenframes on those orbits. Consequently its inclusion does not convert the singlet mass relation into a golden CKM relation. Scalar interactions and mixed higher-loop graphs retain that task.

Exact fermion determinant and counterterms

For each sector let the canonical six-state Dirac matrix have blocks $D = \begin{bmatrix} I & 0 \\ C & m I \end{bmatrix}$, where C is Hermitian, $a^2 = y^2$, $m = m_0 + h$ is real. Direct multiplication for a fully general symbolic Hermitian three-by-three matrix gives $\text{Tr}[(D^\dagger D)^2] = \text{Tr}(C^4) + 2(a^2 + m^2)\text{Tr}(C^2) + 3(a^4 + m^4)$. The identity includes all six Dirac states and holds without a hierarchy expansion or a chosen family frame.

Write $C = z I + g A$ with traceless Hermitian A and $z = e^{\eta + b s}$. Define $N = \text{Tr}(A^2)$, $T = \text{Tr}(A^3)$. Then $\text{Tr}(C^2) = 3z^2 + g^2 N$ and $\text{Tr}(C^4) = 3z^4 + 6z^2 g^2 N + 4z g^3 T + g^4 N^2$. The last term uses the exact three-by-three traceless Cayley-Hamilton identity $\text{Tr}(A^4) = N^2/2$. The driver expands the complete determinant polynomial into 33 monomials per sector, covering field degrees zero through four.

In the declared convention, $16\pi^2 \beta_{\text{box}}(V) = -2 N_c \sum_f \text{Tr}[(D_f^\dagger D_f)^2]$, with $N_c=3$. This is the fermion determinant contribution at fixed renormalized field coordinates. Anomalous dimensions, vertex running, scalar and gauge diagrams are not included in β_{box} . The corresponding finite MS potential is $-N_c/(16\pi^2) \sum_i m_i^4 [\log(m_i^2/\mu^2) - 3/2]$. Differentiating its explicit scale dependence cancels the displayed box contribution. These conventions follow the general one-loop effective potential in Stephen P. Martin, hep-ph/0111209, equations 1.1-1.3 and section 3; the model-specific block identity and operator expansion are derived here.

The new singlet structures are explicit. Per sector the box generates σ^4 with coefficient $-18 h^4$, σ^3 with coefficient $-72 m_0 h$, $\sigma^2 \text{Tr}(C^2)$ with coefficient $-12 h^2$, and $\sigma \text{Tr}(C^2)$ with coefficient $-24 m_0 h$. It also generates the singlet tadpole, mass and vacuum-energy terms, all contained in the complete polynomial. Neither an independent cross-sector angular contraction nor the $\sigma^2 H^\dagger H$ quartic appears in this fermion-box contribution. This exact zero is not a symmetry prohibition.

A scalar loop fills the missing singlet-Higgs portal

Take the allowed quartics $\kappa \sigma^2 N$ and $\rho (H^\dagger H) N$ with eight canonical real adjoint components. Their cross Hessian contraction in $16\pi^2 \beta_{\text{scalar}}(V) = \text{Tr}(\text{Hessian } V^2)/2$ gives $32 \kappa \rho \sigma^2 H^\dagger H$. All off-block contractions vanish; the eight source-source diagonal entries supply the factor 32. The independent test differentiates the complete polynomial on σ , all eight source coordinates and all four real Higgs coordinates.

Thus a zero singlet-Higgs quartic is not retained when both source portals are present. The fermion box also makes κ nonzero for a nonzero singlet heavy Yukawa and adjoint source Yukawa. The coefficient 32 is one specified contribution to the portal beta, rather than a complete beta function or a claim that no other terms can cancel it. This identifies the required operator directly and avoids replacing it by an extra ordinary phase charge.

General classical matching and the induced metric

Let $V(\sigma, q)$ be any real singlet potential depending on the nine source/Higgs singlets q . Select an isolated stationary branch $\sigma_{\text{star}}(q)$ with positive curvature $V_{\sigma\sigma}$. The heavy masses are $M_f(q) = [m_f + h_f \sigma_{\text{star}}(q)]/I$. Therefore $h_{\text{down}}(M_{\text{up}} - m_{\text{up}}/I) - h_{\text{up}}(M_{\text{down}} - m_{\text{down}}/I) = 0$ as an exact functional identity. Every Taylor coefficient shares the same two-component h vector. The relation does not require a Gaussian scalar potential or fixed numerical ratios of the nine source-current coefficients.

Implicit differentiation gives $\partial_i \sigma_{\text{star}} = -V_{\sigma q_i}/V_{\sigma\sigma}$. The local quadratic four-fermion exchange is $(\sum_f h_f F_f)^2/(2 V_{\sigma\sigma})$ in the Lagrangian. Eliminating a canonically normalized singlet adds $\text{grad}_x[\sigma_{\text{star}}(q(x))] \text{grad}_x[\sigma_{\text{star}}(q(x))] - \text{transpose}$ to the kinetic metric, where x are canonical real source/Higgs coordinates, rather than the invariant labels q . This addition is positive semidefinite and has rank at most one; other mediators retain their own metric contributions. At a constant vacuum it produces no quadratic quark kinetic correction. Higher-derivative and multi-fermion terms follow from the same classical elimination; finite loop matching remains distinct.

For $V_{\sigma} = (M^2 + K(q)) \sigma + r_3 \sigma^2 + r_4 \sigma^3 + J(q)$, the small-source branch through quadratic order is $\sigma_{\text{star}} = -J/M^2 + J K/M^4 - r_3 J^2/M^6$. The exact receipt checks the stationary equation order by order. A non-Gaussian numerical branch independently checks the implicit derivative, positive metric and cross-sector rank-one response.

At any such background the quadratic scalar pole problem is equally general. With source Hessian block A , source-singlet mixing column D and positive singlet curvature c , the exact Schur inverse propagator is $A - p^2 I - D D^{\text{transpose}}/(c - p^2)$. The reduced zero-momentum Hessian is $H = A - D D^{\text{transpose}}/c$. Consequently the pole metric is $I + D D^{\text{transpose}}/[c(c - p^2)]$, whose zero-momentum value agrees with the implicit-branch metric. This is quadratic finite-momentum matching, not an expansion that requires the singlet potential to remain Gaussian.

An explicit bounded extension preserving the local benchmark

Let $t = \sigma - \sigma_{\text{vacuum}}$ and q_0 be the nine invariant values of the retained vacuum. Add $r_3 t^3/3 + r_4 t^4/4 + t^2 \sum_i k_i (q_i - q_{0i})/2$. Choose $r_3 = 7/100$, $r_4 = 9/100$ and $k = (0.03, 0.04, 0.005, 0.02, 0.035, 0.045, 0.004, 0.025, 0.05)$, in the established source/Higgs operator order. Both η/s quadratic forms are positive definite; the two adjoint-norm and Higgs coefficients are positive. Every singlet quadratic portal is nonzero, including the singlet-Higgs term. The cubic and quartic self coefficients are nonzero as well.

The added value, gradient and Hessian vanish exactly at $t=0, q=q_0$. The full 49-scalar Hessian, its 21 light poles, the canonical quark matrices and all neutral-scalar quark vertices are therefore unchanged. The preceding positive local Hessian, physical quartet J approximately 0.0002620466 and first-order neutral-scalar phase cancellation carry over by this exact local-jet identity. They are not recomputed as an independent new benchmark. The positive t^4 term bounds the cubic and the subtracted quadratic from below; the positive mixed quartics and prior bounded source/Gaussian-square potential establish boundedness of the extension.

These centered contact choices preserve a chosen local branch. They are not an RG-invariant equality among potential coefficients, and they do not select a global CP-breaking vacuum. General running can move that branch. The representation and fermion-vertex protection argument remains applicable wherever a stable positive-mass branch survives.

Exact orientation neutrality and reproduction

Under C to $U C U^\dagger$, D transforms by the unitary similarity $\text{diag}(U, U)$. All six Dirac singular masses remain identical. Each of the nine singlet source currents is also invariant under independent source conjugations. Thus both the full finite fermion determinant and its divergent polynomial are orientation independent along fixed-eigenvalue orbits. No extraction of a phase convention is involved.

Eight executable source rotations of the singlet-matched benchmark change the physical charged-current quartet across approximately -0.00208045 to $+0.00128810$ while preserving the masses, singlet current and one-loop fermion potential to floating-point precision. The maximum energy discrepancy is approximately 5.6 times ten to the minus seventeen. These orbits are not asserted to be stationary points of one fixed finite-group potential. The exact similarity identity explains the quantum statement; the numerical orbit calculation demonstrates its connection to physical current observables.

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_flavor_singlet_quantum.py` to reproduce the scientific receipt. The focused tests cover the symbolic identity, random independent block traces, scale normalization, generated coefficients, full scalar portal Hessians, non-Gaussian matching, the preserved vacuum jet, canonical metric and physical fixed-spectrum orbits. The result supplies the required quantum-generated scalar content and extends protected classical matching. A correlated frame potential and complete finite gauge/Yukawa/EFT matching remain separate physical obligations.